Switch Selection and PoE Research

Unmanaged Switch:

Brand and Model: NETGEAR 8-port 2.5 Multi-Gigabit Ethernet Unmanaged Network Switch

Price: $129.99

Number of ports: 8

It's not PoE

I think this switch is used for connecting multiple devices to a PC.

This switch looks like its more for a business setting given that it's on the more expensive side of switch's.

https://www.amazon.com/NETGEAR-8-Port-2-5GB-Switch-Multi-Gigabit/dp/B0D54STHFZ/ref=sr_1_2_sspa?crid=16695XMCXNM0E&dib=eyJ2IjoiMSJ9.8XWMINo0GEv4Jr_4axIzQ-YCVWJ350F_0JHmkDaPevi9OMo6JXPD3hwfwxvTlQ0lN5kC1RUy7yH_uftM8lGT6ZgfegwatIglk2_9y_t0nkhAHSpCLC5yIZRX0n_XdE0-wR2KnHqFMc5KYT7pvAZU-rereyAQNCr1sw6GTJiCq-qpR0OONYJ-DjfyoqctH1EnbOIQ5f-yRAOL6tQ-Y5w9KT_CJMUxXmnoutZNWQqH_1STz6xE_Zb35ZLYc_FpQhOFopqGJrtGhzmDPGxZZ8fMqyhQ4I3Yfk7g22EbXE-HgZ0.a8WpVHBeU9rZ4nnHyXfsFfCyPJ74vo-EGYzOvCjWIh0&dib_tag=se&keywords=unmanaged%2Bswitch&qid=1790811425&s=electronics&sprefix=unmanaged%2Bswitch%2Celectronics%2C168&sr=1-2-spons&sp_csd=d2lkZ2V0TmFtZT1zcF9hdGY&th=1

—



Managed Switch:

NETGEAR

$59.99

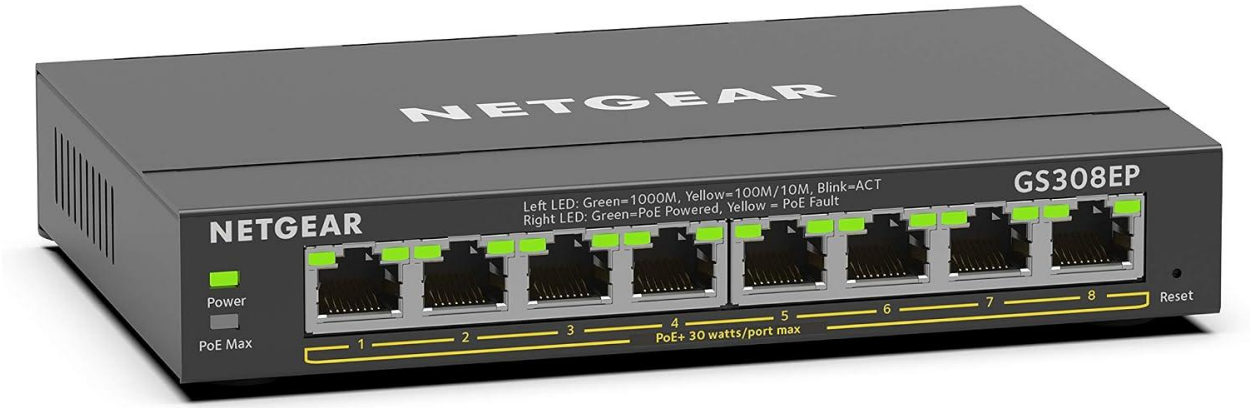
8 ports

No it doesn’t support VLANs

Its compatible with IP cameras and VoIP phones. Maximum power is 62 watts and has 2 volts. It transfers 16 gigabits per second.

This would be used in a business setting. The description says it has advanced network management feature to help optimize performance. It could be used for connecting multiple computers to a router.

https://www.amazon.com/NETGEAR-Gigabit-Ethernet-Switch-GS308EP/dp/B08MBFLMDC/ref=sr_1_1_sspa?dib=eyJ2IjoiMSJ9.1AB3pUuTPZLsDJD1pry1Ocgdxr3Ml_XhUngJzKLWKG-J9ZwQNGfedw8TEwez6ldQ-naQ3hMHRq1Gv4z_rO0Acw9jDW-lFJp-r8DWKiUI5z2CQszsdnzfnByyzfHKDPcm-8smsr3wFX-iMbUYL--pN1MXHPpJ6rOXVXU7Hn3A3o-tpy594pbx_zqiB3S7c_ZrQHcehiVCTbgY1YDg2E1dVjASo40RxHzBoaL4lEfsmjg.4hn0PV7JX8IADAa1Ofd2ihd0yosC6_EL6R4ev4pS2n0&dib_tag=se&keywords=Managed%2BSwitch&qid=1790813462&sr=8-1-spons&sp_csd=d2lkZ2V0TmFtZT1zcF9hdGY&th=1



PoE Switch:

NETGEAR

$54.99

5 ports

Just PoE

Desktop PC

PoE is useful because it reduces the amount of cables someone has to use to get the same result if they weren't using a PoE switch. This also allows more devices to connect since there are more open ports.

1. What is the main difference between a managed and unmanaged switch?

Control is the main difference. A managed switch allows admin to configure, monitor and manage network traffic. Managed switches support VLANs and remote management where as unmanaged just lets devices connect to the network.

2. When would you choose a PoE switch over a regular switch?

I would choose a PoE switch when I need to put up a camera or something that having less cords is the most optimal for it. I would choose to use a regular switch when I want a wired connection for example connecting an ethernet cable to my new Xbox and my Old Xbox.

3. Why would a business need a managed switch instead of unmanaged?

Using a managed switch is important in a business setting because they are more secure and the admin can monitor the network traffic. A managed switch also prioritizes the traffic going through it so important programs can have priority over the less important programs.

4. What is one downside of PoE switches?

PoE switches cost more then regular switch's do. That means if one goes out, they're also expensive to replace. They can also overheat if you connect too many devices that use a lot of power to them for extended periods of time.

My small business scenario:

The switch I would use in my business would be the TP-Link TL-SG1016PE. It's a 16-port Gigabit PoE+ switch, the price is $149.99.

I would choose managed so we could set up VLANs to keep guest Wi-Fi and security footage separate from other data. Using this we could also configure quality of service to not slow down network speeds.

I would choose PoE so cable management isn't a nightmare, as well we'd be able to have security camera's and network connections. All around the cable management would be the best with a PoE switch.

In order to only use one switch, PoE is the best option because 10 employees, 2 wireless access points and 4 security cameras. A 16-port switch can cover all of that. A managed switch is needed to logically separate the camera footage and Wi-Fi from office traffic. Choosing a PoE switch would help with cable management and installation. For $150 all of the use that we'd get out of it is definitely worth it for the security and efficiency.